

OPERATING SYSTEMS PSET 1 REPORT

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1. Sleep.c

- a. Sleep with 50 ticks

```
$ sleep 50
```

```
$ $ $ $ |
```

- b. Sleep without ticks (testing)

```
$ sleep
Usage: sleep <ticks>
```

- c. Using commands during sleep execution

```
$ sleep 50
anis
29017
$ exec anis failed
$ exec 29017 failed
```

- d. grade of sleep

```
anis@Anis:~/xv6-labs-2025$ ./grade-lab-util sleep
make: 'kernel/kernel' is up to date.
== Test sleep, no arguments == sleep, no arguments: OK (2.3s)
== Test sleep, returns == sleep, returns: OK (0.7s)
== Test sleep, makes syscall == sleep, makes syscall: OK (0.8s)
```

2. Sixfive.c

- a. Creating a file

```
$ echo "5 hello 100 18 six /6, xv6" > sixfive.txt
```

- b. Running sixfive on the file and getting the expected output

```
$ sixfive sixfive.txt
5
100
18
6
```

- c. Grade of sixfive

```
anis@Anis:~/xv6-labs-2025$ ./grade-lab-util sixfive
make: 'kernel/kernel' is up to date.
== Test sixfive_test == sixfive_test: OK (1.6s)
== Test sixfive_readme == sixfive_readme: OK (0.9s)
== Test sixfive_all == sixfive_all: OK (1.0s)
```

3. Find.c

- a. Find without a parameter

```
$ $ $ $ find /  
usage: find <start-path> <name> [-exec <cmd> [args...]]
```

- b. Find with sh

```
$ find / sh  
//sh
```

- c. Find with ls

```
$ find . ls  
./ls  
$ This is Anis 29017
```

- d. Expected output of find

```
$ echo > b  
$ mkdir a  
$ echo > a/b  
$ mkdir a/aa  
$ echo > a/aa/b  
$ find . b  
./b  
./a/b  
./a/aa/b  
$
```

- e. Grade of find

```
anis@Anis:~/xv6-labs-2025$ ./grade-lab-util find  
make: 'kernel/kernel' is up to date.  
== Test find, in current directory == find, in current directory: OK (1.5s)  
== Test find, in sub-directory == find, in sub-directory: OK (1.1s)  
== Test find, recursive == find, recursive: OK (1.5s)  
== Test exec, recursive find == exec, recursive find: OK (1.8s)
```

4. Memdump.c

- a. Output without argument

```

$ memdump
Example 1:
61810
2025
Example 2:
a string
Example 3:
another
Example 4:
c88
1819438967
100
z
xyzyy
Example 5:
hello
w
o
r
l
d
#

```

b. Output with argument

```

$ echo deadc0de | memdump hhcccc
25956
25697
c
0
d
e
$ echo deadc0de | memdump p
64616564

```

c. Grade of memdump

```

anis@Anis:~/xv6-labs-2025$ ./grade-lab-util memdump
make: 'kernel/kernel' is up to date.
== Test memdump, examples == memdump, examples: OK (0.6s)
== Test memdump, format ii, S, p == memdump, format ii, S, p: OK (0.9s)

```

5. Exec.c

a. Output of example

```
$ find . wc -exec echo hi  
hi ./wc
```

b. Expected output

```
$ sh < findtest.sh  
$ $ $ $ $ $ hello  
hello  
hello  
$ $
```

c. Grade of exec

```
make: 'kernel/kernel' is up to date.  
== Test exec == exec: OK (0.9s)  
== Test exec, multiple args == exec, multiple args: OK (1.0s)  
== Test exec, recursive find == exec, recursive find: OK (1.1s)
```